A PROPERTY OF A COMPOSITION SURVIVES ALONG A SURJECTION OF SHEAVES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites three Morphisms of Spaces definitions and one Descent lemma, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Definitions of flat, locally finite type, locally finite presentation for algebraic-space morphisms. Stacks Project, Morphisms of Spaces, Definitions spaces-morphisms-definition-flat, spaces-morphisms-definition-locally-finite-type, spaces-morphisms-definition-locally-finite-presentation. Reduce the property to the chart morphisms $U \to W$ and $V \to W$.
- Scheme case. Stacks Project, Descent, Lemma descent-lemma-curiosity. Concludes for $V \to W$.

Source reference. Original reference 03ML: sources/stacks-chow/tags.txt, source lines 4753–4753. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03XF: sources/stacks-chow/tags.txt, source lines 5146–5146. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03XP: sources/stacks-chow/tags.txt, source lines 5156–5156. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 06NB: sources/stacks-chow/tags.txt, source lines 8530–8530. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let $X \to Y \to Z$ be morphism of algebraic spaces. Let P be one of the following properties of morphisms of algebraic spaces over S: flat, locally finite type, locally finite presentation. Assume that $X \to Z$ has P and that $X \to Y$ is a surjection of sheaves on $(Sch/S)_{fppf}$. Then $Y \to Z$ is P.*

Proof. Choose a scheme W and a surjective étale morphism $W \to Z$. Choose a scheme V and a surjective étale morphism $V \to W \times_Z Y$. Choose a scheme U and a surjective étale morphism $U \to V \times_Y X$. By assumption we can find an fppf covering $\{V_i \to V\}$ and lifts $V_i \to X$ of the morphism $V_i \to Y$. Since $U \to X$ is surjective étale we see that over the members of the fppf covering $\{V_i \times_X U \to V\}$ we have lifts into U. Hence $U \to V$ induces a surjection of sheaves on $(Sch/S)_{fppf}$. By our definition of what it means to have property P for a morphism of algebraic spaces (see Morphisms of Spaces, Definition 03ML, Definition 03XF, and Definition

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03XP) we see that $U \to W$ has P and we have to show $V \to W$ has P. Thus we reduce the question to the case of morphisms of schemes which is treated in Descent, Lemma 06NB. $\square$